Non-uniform Non-Archimedean Estimate for Regular Domains

Statement

Let $(X, \mu, \mathcal{P})$ be a good grid and let

$$0 < s < \frac{1}{p}, \qquad 1 \le p < \infty, 1 \le q \le infty$$

For $Q \in \mathcal{P}^k$, write $|Q| = \mu(Q)$, and use the Souza atom

$$\psi_Q = |Q|^{s-\frac{1}{p}} \mathbf{1}_Q.$$

There is $C_{na} \ge 0$ such that the following holds: Let $f \in \mathcal{B}^s_{p,q} \cap L^\infty$ be represented

$$f = \sum_{k \ge 0} \sum_{Q \in \mathcal{P}^k} c_Q \psi_Q,$$

by either an standard representation or a positive repsentation, with coefficient finite cost

$$\|c\|_{b^s_{p,q}} = \begin{cases} \left(\sum_{k \ge 0} \left(\sum_{Q \in \mathcal{P}^k} |c_Q|^p \right)^{q/p} \right)^{1/q}, & q < \infty, \\ \sup_{k \ge 0} \left(\sum_{Q \in \mathcal{P}^k} |c_Q|^p \right)^{1/p}, & q = \infty. \end{cases}$$

Assume also that

$$|f(x)| \le M \qquad \text{for a.e. } x.$$

Let $(\Omega_i)_{i \in \Lambda}$ be pairwise disjoint domains, such that 1_{Ω_i} with a positive representation

$$\mathbf{1}_{\Omega_i} = \sum_{k \ge 0} \sum_{Q \in \mathcal{P}^k} b^i_Q \psi_Q$$

such that

$$b^i_Q \ne 0 \implies b^i_Q > 0 \implies Q \subset \Omega_i, \qquad \left| b^i_Q \right| \le \mathcal{C}_i,$$

and such that $\|c\|_{b^s_{p,q}} \le \mathcal{C}_i$

Let $(\Theta_i)_{i \in \Lambda}$ be real weights. The main hypothesis is that, for every active source cell P, that is, for every $P \in \mathcal{P}$ with $c_P \ne 0$,

$$\sum_{\substack{i \in \Lambda \\ P \cap \Omega_i \ne \emptyset}} |\Theta_i| \left(1 + \mathcal{C}_i\right) \le N. \tag{1}$$

Define

$$h = \sum_{i \in \Lambda} \Theta_i \mathbf{1}_{\Omega_i} f.$$

Then there is a representation S of h, supported in the domains Ω_i, such that

$$\mathrm{pqCost}(S) + \|h\|_{L^\infty} \le C_{\mathrm{na}}\, N\,\big(\mathrm{pqCost}(R) + M\big), \tag{2}$$

where $R = (c_Q)_Q$ is the chosen representation of f, and C_{na} depends only on the structural constants of the grid and on s, p, q. Moreover, if a coefficient of S at a cell Q is nonzero, then

$$Q \subset \Omega_i \qquad \text{for some } i \in \Lambda.$$

The Product Decomposition

Suppose first that Λ is finite. If

$$\mathbf{1}_{\Omega_i} = \sum_Q b^i_Q \psi_Q \qquad \text{and} \qquad f = \sum_P c_P \psi_P.$$

The product is decomposed into two parts:

$$\left(\sum_{i\in\Lambda} \Theta_i \mathbf{1}_{\Omega_i}\right) f = \sum_{i\in\Lambda} U_{1,i} + \sum_{i\in\Lambda} U_{2,i}.$$

The first term $U_{1,i}$ collects the interactions where the indicator cell is below the source cell. If $Q \in \mathcal{P}^j$, its coefficient is

$$(U_{1,i})_Q = \Theta_i b^i_Q\, \mathcal{T}_R(Q), \tag{3}$$

where

$$\mathcal{T}_R(Q) = \sum_{\ell\le j} \sum_{\substack{P\in\mathcal{P}^\ell \\ Q\subset P}} c_P\, |P|^{s-\frac{1}{p}}. \tag{4}$$

Since the representation of f is either standard or positive we have

$$\|\mathcal{T}_R(Q)\| \le M.$$

This is the ancestral tower of the source representation R, evaluated on the cell Q. In Lean this is

$$\mathrm{weightedAncestorCoeffSum}(G, R, Q).$$

The second term $U_{2,i}$ collects the complementary interactions, where the source cell is below the indicator cell. Its coefficients have the schematic form

$$(U_{2,i})_J = \Theta_i c_J\, \mathcal{T}^<_i(J), \tag{5}$$

where $\mathcal{T}^<_i(J)$ is the strict ancestor tower of the indicator representation above J. More explicitly, if $J \in \mathcal{P}^j$, then

$$\mathcal{T}^<_i(J) = \sum_{\ell<j} \sum_{\substack{Q\in\mathcal{P}^\ell \\ J\subset Q}} b^i_Q\, |Q|^{s-\frac{1}{p}}. \tag{6}$$

The word "strict" means that the level $\ell = j$ is excluded. Since this representation is positive we have

$$\|\mathcal{T}^<_i(J)\| \le \|1_{\Omega_i}\|_\infty \le 1.$$

In the Lean formalization this is

$$\mathrm{strictWeightedAncestorCoeffSum}(G, R_i, J),$$

where R_i is the chosen indicator representation of $\mathbf{1}_{\Omega_i}$.

Define

$$U_1 = \sum_{i\in\Lambda} U_{1,i}$$

and

$$U_2 = \sum_{i\in\Lambda} U_{2,i}$$

We have

$$(U_1)_Q = \sum_{i\in\Lambda} \Theta_i b^i_Q \, \mathcal{T}_R(Q),$$

$$(U_2)_J = \sum_{i\in\Lambda} \Theta_i c_J \, \mathcal{T}^<_i(J),$$

The U_1 estimate

The cost at level k of U_1. If $b^i_Q \neq 0$ for some i then $Q \subset \Omega_i$ and we define $i(Q) := i$. Otherwise define $i(Q)$ as an arbitrary $i \in \Lambda$. If $\mathcal{T}_R(Q) \neq 0$ and $Q \subset \Omega_i$ then there is $Q \subset P$ such that $c_P \neq 0$ so $P \cap \Omega_{i(Q)} \neq \emptyset$, and in particular $\|\Theta_{i(Q)}\| \leq N$.

$$\left(\sum_{Q\in\mathcal{P}^k} \| \sum_{i\in\Lambda} \Theta_i b^i_Q \, \mathcal{T}_R(Q)\|^p\right)^{1/p} \tag{7}$$

$$\leq \left(\sum_{Q\in\mathcal{P}^k} \|\Theta_{i(Q)} b^{i(Q)}_Q \, \mathcal{T}_R(Q)\|^p\right)^{1/p} \tag{8}$$

$$\leq M \left(\sum_{\substack{Q\in\mathcal{P}^k \\ \mathcal{T}_R(Q)\neq 0}} \|\Theta_{i(Q)} b^{i(Q)}_Q\|^p\right)^{1/p} \tag{9}$$

$$\leq MN \left(\sum_{Q\in\mathcal{P}^k} \|b^{i(Q)}_Q\|^p\right)^{1/p} \tag{10}$$

$$\tag{11}$$

so

$$\left(\sum_k \left(\sum_{Q \in \mathcal{P}^k} \| \sum_{i \in \Lambda} b_Q^i \, \mathcal{T}_R(Q) \|^p \right)^{q/p} \right)^{1/q} \tag{12}$$

$$\leq M \left(\sum_k \left(\sum_{\imath \in \Lambda} \|\Theta_i\| \left(\sum_{Q \in \mathcal{P}^k} \|b_Q^i\|^p \right)^{1/p} \right)^q \right)^{1/q} \tag{13}$$

$$\leq M \sum_{\imath \in \Lambda} \|\Theta_i\| \left(\sum_k \left(\sum_{Q \in \mathcal{P}^k} \|b_Q^i\|^p \right)^{q/p} \right)^{1/q} \tag{14}$$

$$\leq M \sum_{i \in \Lambda} \|\Theta_i\| \mathcal{C}_i. \tag{15}$$

For a fixed $Q \in \mathcal{P}^j$, suppose

$$\Theta_i(U_{1,i})_Q \neq 0.$$

Then $b_Q^i \neq 0$, hence $Q \subset \Omega_i$. Also, if $\mathcal{T}_R(Q) \neq 0$, then some ancestor $P \supset Q$ has $c_P \neq 0$. Thus $P \cap \Omega_i \neq \emptyset$. Applying the overlap hypothesis (1) to this active cell P gives

$$|\Theta_i| \, (1 + \mathcal{C}_i) \leq N.$$

Since $\left| b_Q^i \right| \leq \mathcal{C}_i \leq 1 + \mathcal{C}_i$, we obtain the local bound

$$|\Theta_i| \left| b_Q^i \right| \leq N. \tag{16}$$

Using (3),

$$|\Theta_i(U_{1,i})_Q| = |\Theta_i| \left| b_Q^i \right| |\mathcal{T}_R(Q)| \leq N \, |\mathcal{T}_R(Q)| \, .$$

Now use disjointness of the domains. For a fixed Q, at most one index i can have $b_Q^i \neq 0$, because $b_Q^i \neq 0$ implies $Q \subset \Omega_i$, and the Ω_i's are pairwise disjoint. Therefore, for every finite $\Gamma \subset \Lambda$,

$$\begin{aligned} \sum_{i \in \Gamma} |\Theta_i|^p \sum_{Q \in \mathcal{P}^j} |(U_{1,i})_Q|^p &= \sum_{Q \in \mathcal{P}^j} \sum_{i \in \Gamma} |\Theta_i(U_{1,i})_Q|^p \\ &\leq \sum_{Q \in \mathcal{P}^j} N^p \, |\mathcal{T}_R(Q)|^p \\ &= N^p T_j(R)^p. \end{aligned} \tag{17}$$

The U_2 estimate

The U_2 term is controlled by the source coefficients themselves. Indeed, the strict indicator tower $\mathcal{T}_i^<(J)$ is nonzero only when an ancestor of J belongs to the canonical decomposition of Ω_i. Thus, if $\Theta_i(U_{2,i})_J \neq 0$, then some source cell meeting the corresponding indicator ancestor also meets Ω_i, and the overlap hypothesis again gives

$$|\Theta_i| \, (1 + \mathcal{C}_i) \leq N.$$

The canonical indicator construction gives

$$\left| \mathcal{T}_i^<(J) \right| \leq 1 + \mathcal{C}_i.$$

Hence

$$|\Theta_i(U_{2,i})_J| \le N\,|c_J|\,.$$

As before, disjointness implies that for each J at most one i may contribute. Consequently,

$$\begin{aligned}\sum_{i\in\Gamma}|\Theta_i|^p \sum_{J\in\mathcal{P}^j}|(U_{2,i})_J|^p &\le N^p \sum_{J\in\mathcal{P}^j}|c_J|^p \\ &= N^p A_j(R)^p. \end{aligned} \tag{18}$$

Combining U_1 and U_2

The product representation for the finite partial sum

$$h_\Gamma = \sum_{i\in\Gamma}\Theta_i \mathbf{1}_{\Omega_i} f$$

has level j coefficient power bounded by the weighted sum of $U_{1,i}+U_{2,i}$. Using

$$|x+y|^p \le 2^{p-1}\big(|x|^p+|y|^p\big),$$

together with (17) and (18), we get

$$\sum_{i\in\Gamma}|\Theta_i|^p \sum_{Q\in\mathcal{P}^j}|(U_{1,i}+U_{2,i})_Q|^p \le 2^{p-1}N^p\big(T_j(R)^p + A_j(R)^p\big). \tag{19}$$

This is the non-uniform level-by-level estimate. Notice that the factor $|\Theta_i|\,\mathcal{C}_i$ is used locally before summing in Q; it is not replaced by a global counting estimate.

The Tower Estimate

The remaining analytic input is the standard tower estimate for canonical Souza representations:

$$\begin{cases} \left(\displaystyle\sum_{j\ge 0} T_j(R)^q\right)^{1/q} \le C_{\text{tow}}\big(\text{pqCost}(R)+M\big), & q<\infty, \\ \displaystyle\sup_{j\ge 0} T_j(R) \le C_{\text{tow}}\big(\text{pqCost}(R)+M\big), & q=\infty. \end{cases} \tag{20}$$

This is the paper-level form of the "tower term". It is a discrete Hardy estimate along the ancestor chains of the grid. The point of (20) is that one controls the whole sequence $(T_j(R))_j$, not merely the pointwise bound $|\mathcal{T}_R(Q)| \le M$. The latter would introduce a false factor counting the number of cells on level j.

Mixed Cost Estimate

Let S_Γ be the representation of h_Γ obtained from the above product blocks. From (19),

$$\text{levelCoeffPower}_j(S_\Gamma) \le 2^{p-1}N^p\big(T_j(R)^p + A_j(R)^p\big).$$

Taking p-roots gives

$$\text{levelCoeffPower}_j(S_\Gamma)^{1/p} \le 2^{1-\frac{1}{p}}N\big(T_j(R)^p + A_j(R)^p\big)^{1/p}.$$

Since

$$\big(a^p+b^p\big)^{1/p} \le a+b \qquad (a,b\ge 0),$$

we obtain

$$\text{levelCoeffPower}_j(S_\Gamma)^{1/p} \le 2^{1-\frac{1}{p}} N\big(T_j(R) + A_j(R)\big).$$

Taking the ℓ^q norm in j, using Minkowski's inequality and (20),

$$\begin{aligned}\text{pqCost}(S_\Gamma) &\le 2^{1-\frac{1}{p}} N \left(\|(T_j(R))_j\|_{\ell^q} + \|(A_j(R))_j\|_{\ell^q}\right)\\ &\le C_1 N\big(\text{pqCost}(R) + M\big),\end{aligned} \tag{21}$$

where C_1 depends only on s, p, q and the grid constants.

Pointwise and L^∞ Estimates

For a fixed point x, pairwise disjointness of the domains implies that at most one indicator $\mathbf{1}_{\Omega_i}(x)$ is nonzero. If $f(x) \ne 0$, choose an active source cell P containing x. Then every i with $\mathbf{1}_{\Omega_i}(x) \ne 0$ satisfies $P \cap \Omega_i \ne \emptyset$. Hence (1) gives

$$\sum_{i\in\Lambda} |\Theta_i \mathbf{1}_{\Omega_i}(x)| \le N.$$

Therefore

$$|h(x)| = \left|\sum_{i\in\Lambda} \Theta_i \mathbf{1}_{\Omega_i}(x) f(x)\right| \le N\,|f(x)| \le NM$$

for a.e. x. Thus

$$\|h\|_{L^\infty} \le NM. \tag{22}$$

Passage to the Infinite Sum

Let $(\Gamma_n)_n$ be an increasing sequence of finite subsets of Λ exhausting Λ, and set

$$h_n = \sum_{i\in\Gamma_n} \Theta_i \mathbf{1}_{\Omega_i} f.$$

The estimates (21) and (22) are uniform in n. By the compactness/limit theorem for bounded families of Besov representations, there is a subsequence of the representations S_{Γ_n} converging to a representation S of

$$h = \sum_{i\in\Lambda} \Theta_i \mathbf{1}_{\Omega_i} f$$

with

$$\text{pqCost}(S) \le C_1 N\big(\text{pqCost}(R) + M\big).$$

The support condition is closed under this limit: if a coefficient of S on a cell Q is nonzero, then $Q \subset \Omega_i$ for some $i \in \Lambda$. Combining this with (22), and increasing the constant if necessary, gives

$$\text{pqCost}(S) + \|h\|_{L^\infty} \le C_{\text{na}} N\big(\text{pqCost}(R) + M\big).$$

This proves (2).

Remark 1. *In the Lean development, the already formalized level estimate is exactly* (19)*. The term*

$$T_j(R)^p = \sum_{Q\in\mathcal{P}^j} |\text{weightedAncestorCoeffSum}(G, R, Q)|^p$$

is the tower term. The remaining formal analytic step is the mixed ℓ^q estimate (20)*; replacing $T_j(R)$ by the pointwise bound M times the number of level cells is not the desired argument.*